

3. (a) Describe a method to investigate the force-extension properties of rubber in the form of an elastic band **as it is loaded**. You should describe how the extension of the rubber is accurately measured. [3]

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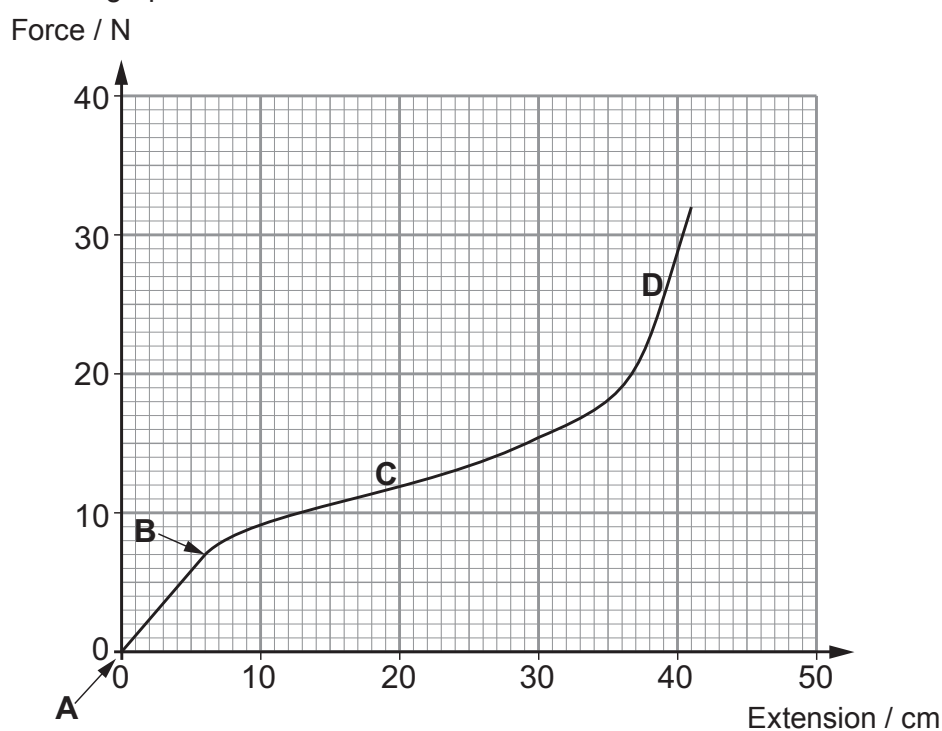
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- (b) The results from such an experiment for a rubber band of unstretched length 8.0 cm are plotted in a graph.



- (i) Calculate the strain in the rubber at point **B**. [1]

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- (ii) Determine the Young modulus of the rubber in the region **AB**. Assume the band has a total cross-sectional area of 0.050 cm^2 . [3]

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- (c) By referring to the molecular structure of rubber, explain why the gradient at **C** is less than the gradient at **D**. [3]

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